

1 Framed disk bundles (?)

Robin is not working on this section for now.

2 Defining the bracket bundle

Robin is not currently working on this section.

3 The Strata of $\tilde{C}_A$

For a finite set S , denote by $|S|$ the number of elements in S .

Let A be a finite set. In this section we describe the Strata of $\tilde{C}_A$.

3.1 The combinatorics

Definition 3.1. An A -labeled tree consists of

- a tree T , meaning a connected acyclic graph, with vertex set $V(T)$ and edge set $E(T)$;
- a distinguished vertex $r \in V(T)$ called the *root* (which may have any positive valence);
- a map $ls : V(T) \rightarrow \{\mathbb{R}^d, \pi_1, \pi_2, \pi_{\hat{f}}\}$, where we call $ls(v)$ the space label of v ; and
- a bijection to A from the set of *leaves* v of T such that $ls(v) = \mathbb{R}^d$, where a leaf is a non-root vertex of valence 1. Thus such leaves are A -labeled, and we have an injective map from the set of all leaves $L(T)$ to $A \sqcup \{\pi_1, \pi_2, \pi_{\hat{f}}\}$.
- We set further terminology: For two distinct vertices $v, w \in V(T)$, if the unique path from r to v passes through w , we say v lies *above* w or is an *ancestor* of w , and we write $v > w$, with $<$ defining a partial order on $V(T)$. If v and w are also adjacent, then we say w is the *parent* of v and v is a *child* of w , and we denote the set of children of v by $Ch(v)$. For any $e \in E(T)$, we say the vertex adjacent to e and closer to r is *just below* e and denote it by $v_-(e)$, and we say the vertex adjacent to e and further away from r is *just above* e and denote it by $v_+(e)$. We view an edge e as directed from $v_-(e)$ to $v_+(e)$.

We require the following conditions:

- if $v \in V(T)$ is such that $ls(v) = \mathbb{R}^d$, then $|Ch(v)| \neq 1$; and
- either both π_1, π_2 are associated to exactly one vertex, and no vertex is associated to $\pi_{\hat{f}}$; or $\pi_{\hat{f}}$ is associated to exactly one vertex, and no vertex is associated to π_1 or π_2 .

Denote by $\mathcal{T}(A)$ the set of A -labeled trees.

As a rooted tree, any A -tree T has the property that the map $e \mapsto v_+(e)$ is a bijection from edges of T to non-root vertices of T .

The strata of $\tilde{C}_A$ will be labeled by elements of $\mathcal{T}(A)$. Our A -labeled trees are similar to the f -trees of Sinha [?] but adapted to allow configuration points in the three bundles under consideration rather than just in $\mathbb{R}^d$. One may also think of an A -labeled leaf as a marked point and any other non-root vertex v (or the edge e such that $v_+(e) = v$) as a node. (Robin: more specific comparison here?)

(In this example, $A = \{1, 2, \dots, 14\}$; the point labels are green and space labels are red. When a space label is $\mathbb{R}^d$, it is omitted. (We probably need to redraw the picture that was here.))

Definition 3.2. We define an *addition* (or *sum*) operation on $\{\mathbb{R}^d, \pi_1, \pi_2, \pi_{\hat{j}}\}$ as the commutative binary operation $+$ such that $\mathbb{R}^d + X = X$, $\pi_1 + \pi_2 = \pi_{\hat{j}}$, and $\pi_{\hat{j}} + X = \pi_{\hat{j}}$ for all $X \in \{\mathbb{R}^d, \pi_1, \pi_2, \pi_{\hat{j}}\}$.

One may view this operation as union on the set of subsets of $\{1, 2\}$, by the identifications $\mathbb{R}^d \mapsto \emptyset$, $\pi_i \mapsto \{i\}$ for $i = 1$ or 2 , and $\pi_{\hat{j}} \mapsto \{1, 2\}$.

Definition 3.3. Let $E^c(T)$ be the set of edges not incident to any A -labeled leaf; we call these *contractible edges*. If $T \in \mathcal{T}(A)$ and $e \in E^c(T)$ is, let $T/e \in \mathcal{T}(A)$ be the tree obtained by contracting e , i.e., merging the two vertices connected to e into a single vertex and taking the sum (in the sense above) of their space labels.

We denote the new vertex in $V(T/e)$ by $[e]$. From the definition of T/e , there are obvious maps

$$\text{contr}_{T;e}^V : V(T) \longrightarrow V(T/e)$$

which maps $v_-(e), v_+(e)$ to $[e]$ and every other vertex to itself, and

$$\text{contr}_{T;e}^E : E(T) \setminus \{e\} \longrightarrow E(T/e)$$

which maps every edge to itself.

If we consider a category whose object set is $\mathcal{T}(A)$ and in which a morphism is an edge contraction, then its final object is the tree with just a root and $|A|$ leaves, with $ls(v) = \mathbb{R}^d$ for every leaf v and $ls(r) = \pi_{\hat{j}}$.

3.2 Describing each stratum

Let T be an A -labeled tree. For $v \in V(T)$, define

$$\text{Space}(v) = C_{Ch(v)}(ls(v)),$$

i.e., the (uncompactified) configuration space of $Ch(v)$ -labeled points in $ls(v)$. (We probably need to redraw the picture that was here and rewrite the next sentence.) For example, for the vertex v in the picture above,

$$ls(v) = \mathbb{R}^d, lp(v) = \{11, 12\}, cld(v) = \{w, w', w''\},$$

so $\text{Space}(v) = C_{\{11, 12, w, w', w''\}}(\mathbb{R}^d)$.

Define $S_T = \prod_{v \in V(T)} \text{Space}(v)$. We may exclude leaves in the product above because $C_\emptyset(M) = *$ for any space label M . Thus for T the final object as above, $S_T = C_A(\pi_{\dot{I}})$. For any $T \in \mathcal{T}(A)$, it follows from definition that S_T is a smooth manifold; it is also the total space of a smooth fiber bundle over $B_1 \times B_2$.

Example: if $A = \{1, 2, 3, 4\}$ and T is the tree below on the left, then S_T consists of configurations like the picture below on the right. (Need to redraw the tree in this example.)

Define $\tilde{C}_A = \bigsqcup_{T \in \mathcal{T}(A)} S_T$ as a set.

3.3 Normalization and radii for charts

Definition 3.4. Suppose $v \in V(T)$ is such that $ls(v) = \mathbb{R}^d$. The *mass* of $w \in Ch(v)$ is defined as

$$\begin{aligned} & \infty && \text{if } \exists x \geq w \text{ such that } ls(x) \in \{\pi_1, \pi_2, \pi_{\dot{I}}\} \\ & |\{x \in L(T) : x \geq w\}| && \text{otherwise.} \end{aligned}$$

Informally, it is the number of leaves above w , unless a vertex with label other than $\mathbb{R}^d$ lies above w , in which case it is infinity.

For example, if w and v are like this (Robin: I changed x to w in the text above)

then on the v -screen, the marked point 1 has weight 1, the node n has weight ∞ , and the node n' has weight 3.

Write an element $\mathbf{x} \in S_T = \prod_{v \in V(T)} \text{Space}(v)$ as $\mathbf{x} = (\mathbf{x}_v)_{v \in V(T)}$. Recall that for any set S , we defined $C_S(\mathbb{R}^d)$ as the quotient of the configuration space of points in $\mathbb{R}^d$ by translation and scaling.

Definition 3.5. Define the *standard representative* of the equivalence class $\mathbf{x}_v \in C_{Ch(v)}(\mathbb{R}^d)$ as the element $\mathbf{x}'_v \in (\mathbb{R}^d)^{Ch(v)}$ such that

- $0 \in \mathbb{R}^d$ is the center of mass of points in $Ch(v)$, where if 2 elements in $Ch(v)$ have weight ∞ , we define the center of mass to be the mid-point of these 2 elements;
- the scaling is such that

$$1 = \inf\{r \in \mathbb{R}^{>0} \mid \text{all marked points and nodes are inside of } \mathbb{D}^d(r)\}.$$

For example, in the picture above, the center of mass of the v -screen coincides with the node n .

(Definitions 3.6 and 3.7 can be skipped on first reading.)

Given $T \in \mathcal{T}(A)$ and $v \in V(T)$, let $L(\geq v)$ be the set of leaves that are either v or children of v , and define

$$ls(> v) = \{x \in \{\pi_1, \pi_2\} \mid \exists w \in V(T) \text{ with } w > v \text{ and either } ls(w) = x \text{ or } ls(w) = \pi_i\}$$

which is roughly the set of space-labels above v , though $ls(> v) = \{\pi_1, \pi_2\}$ if π_i labels a vertex above v .

Definition 3.6. Suppose $\mathbf{x} = (\mathbf{x}_v)_{v \in V(T)} \in S_T$ and $v \in V(T)$. Define $R^{\min}(\mathbf{x}; v)$ as follows:

- If $ls(v) = \mathbb{R}^d$, then $R^{\min}(\mathbf{x}; v) = 2$.
- If $ls(v) = \pi_i$, $i = 1$ or 2 , and $ls(> v) = \emptyset$, then $R^{\min}(\mathbf{x}; v) =$

$$2 \cdot \inf\{R \in [1, \infty) : \text{in the configuration } \mathbf{x}_v, \text{ all points are inside } E_i(R)\}.$$
- If $ls(v) = \pi_i$, $i = 1$ or 2 , and $ls(> v) = \{\pi_{3-i}\}$, first denote by $\star_v \in E_i - \sigma_i(B_i)$ the position of the point w labeled π_{3-i} in the configuration $\mathbf{x}_v$. Then we can view $\mathbf{x}_v$ as an element in

$$C_{Ch(v) \setminus \{w\}}(\pi_i^* E_i)|_{\star_v},$$

where $\pi_i^* E_i$ is the total space of the tautological bundle as in Section ??, and $(\pi_i^* E_i)|_{\star_v}$ denotes the fiber of $\pi_i^*(E_i)$ over the point $\star_v$. Now define $R^{\min}(x; v) =$

$$2 \cdot \inf\{R \in [3, \infty) : \text{in the configuration } \mathbf{x}_v, \text{ all points are inside } \hat{E}_i(R)\}.$$

- If $ls(v) = \pi_i$, then $R^{\min}(x; v) =$

$$2 \cdot \inf\{R \in [1, \infty) : \text{in the configuration } \mathbf{x}_v, \text{ all points are inside } E_i(R)\}.$$

Definition 3.7. Suppose $\mathbf{x} = (\mathbf{x}_v)_{v \in V(T)} \in S_T$, and $e \in E^c(T)$. Note that $v_+(e) \in Ch(v_-(e))$. If $ls(v_-(e)) \neq \mathbb{R}^d$, we denote by $\star_e \in ls(v_-(e))$ the location of the $v_+(e)$ -marked point in the configuration $\mathbf{x}_{v_-(e)}$; if $ls(v_-(e)) = \mathbb{R}^d$, we denote by $\star_e \in \mathbb{R}^d$ the location of the $v_+(e)$ -marked point in the configuration representative $\mathbf{x}'_{v_-(e)}$. Define $r^{\max}(\mathbf{x}; e)$ as follows:

- If $ls(v_-(e)) = \mathbb{R}^d$, then

$$r^{\max}(\mathbf{x}; e) = \frac{1}{2} \cdot \sup\{r \in (-\infty, 1) : \text{in the representative configuration } \mathbf{x}'_{v_-(e)},$$

$$\text{all points except } \star_e \text{ are outside } \mathbb{D}_{\star_e}(r)\}.$$

- If $ls(v_-(e)) = \pi_i$ or π_i , $i = 1$ or 2 , then

$$r^{\max}(\mathbf{x}; e) = \frac{1}{2} \cdot \sup\{r \in (-\infty, \text{radius}^{\max}(\star_e)) : \text{in the configuration } \mathbf{x}_{v_-(e)},$$

$$\text{all points except } \star_e \text{ are outside } D_{\star_e}(r)\},$$

where $\text{radius}^{\max}(\star_e)$ is as in ?? and $D_{\star_e}(r)$ is as in (??).

Definition 3.8. For all $T \in \mathcal{T}(A)$ and $e \in E^c(T)$, define the *maximal smoothing parameter function*

$$\epsilon_{T;e}^{\max} : S_T \longrightarrow \mathbb{R}^{>0}, \quad \epsilon_{T;e}^{\max}(\mathbf{x}) = r^{\max}(\mathbf{x}; e) / R^{\min}(\mathbf{x}; v_+(e)).$$

It is easy to see that $\epsilon_{T;e}^{\max}$ is smooth. Note that the infimum of $\epsilon_{T;e}^{\max}$ is 0.

4 The manifold structure on $\tilde{C}_A$

Recall that we defined $\tilde{C}_A = \bigsqcup_{T \in \mathcal{T}(A)} S_T$ as a set. In this section we define the smooth-manifold-with-corners structure on $\tilde{C}_A$.

For each $T \in \mathcal{T}(A)$, define the *domain* of T as

$$\text{dom}(T) = \{(\mathbf{x}, (\epsilon_e)_{e \in E^c(T)}) \in S_T \times [0, \infty)^{E^c(T)} : \forall e, \epsilon_e < \epsilon_{T;e}^{\max}\}.$$

For $e \in E^c(T)$, also define

$$\text{dom}(T; e) = \{(\mathbf{x}, \epsilon_e) \in S_T \times [0, \infty) : \epsilon_e < \epsilon_{T;e}^{\max}\}.$$

We will construct injective maps

$$\psi_T : \text{dom}(T) \longrightarrow \tilde{C}_A$$

such that $\psi_T|_{S_T \times (0, \dots, 0)}$ is the identity map on S_T . These maps will serve as smooth charts for $\tilde{C}_A$. After defining these maps, we need to check that the transition maps are smooth.

The process of defining ψ_T is by smoothing out the nodes. We first define how to smooth out one node.

4.1 Smoothing out one node

(Robin: I repeated these first two paragraphs in the Definition below...)

Suppose $T \in \mathcal{T}(A)$ and $e \in E^c(T)$. We construct a map

$$\psi_{T;e} : \text{dom}(T; e) \longrightarrow S_T \sqcup S_{T/e}.$$

Define $\psi_{T;e}|_{S_T \times \{0\}}$ to map by the projection $S_T \times \{0\} \rightarrow S_T$.

If $\mathbf{x} = (\mathbf{x}_v \in C_{Ch(v)}(ls(v)))_{v \in V(T)}$ and $\epsilon > 0$ is small enough so that $(\mathbf{x}, \epsilon) \in \text{dom}(T; e)$, then define

$$\psi_{T;e}(\mathbf{x}, \epsilon) = (\mathbf{y}_v \in C_{Ch(v)}(ls(v)))_{v \in V(T/e)}$$

as follows. If v is not an endpoint of e , put $\mathbf{y}_{\text{contr}_{T;e}^V(v)} = \mathbf{x}_v$. It remains to define

$$\mathbf{y}_{[e]} \in C_{Ch([e])}(ls(v_-(e)) + ls(v_+(e))).$$

Note that $Ch([e]) = Ch(v_+(e)) \sqcup Ch(v_-(e)) \setminus \{v_+(e)\}$.

(Robin: make this a subsection??) Before going into details, we explain intuitively how $\mathbf{y}_{[e]}$ is defined. It is obtained from $\mathbf{x}_{v_-(e)}$ and $\mathbf{x}_{v_+(e)}$ as follows.

Take, arbitrarily, $R > R^{\min}(\mathbf{x}; v_+(e))$ and $r < r^{\max}(\mathbf{x}; e)$ such that $r/R = \epsilon$.

Draw $\mathbf{x}_{v_-(e)}$ and $\mathbf{x}_{v_+(e)}$ as below:

where $\mathbf{x}_{v_-(e)}$ (resp. $\mathbf{x}_{v_+(e)}$) is the configuration of green and blue points on $ls(v_-(e))$ (resp. $ls(v_+(e))$). Here the space where the marked points lie on, denoted by $E_{\pm}$, is the fiber of the bundle $ls(v_{\pm}(e))$ on which $\mathbf{x}_{v_{\pm}(e)}$ lies, if $ls(v_{\pm}(e)) \neq \mathbb{R}^d$; if $ls(v_{\pm}(e)) = \mathbb{R}^d$, then $E_{\pm} = \mathbb{R}^d$, and the positions of points on it are given by $\mathbf{x}'_{v_{\pm}(e)}$.

Next, we cut off a disk of radius r in E_- , centered at $\star_e$. Here, if $E_- \neq \mathbb{R}^d$, “radius r ” is made sense of by the exponential map.

We also cut off a neighborhood of $\star_e$ in E_+ . This neighborhood is the one identified with $\mathbb{R}^d \setminus \mathbb{D}^d(R)$ via a “trivialization at ∞ ” of $ls(v_+(e))$. This “trivialization at ∞ ” depends on $ls(v_+(e))$ and $ls(> v_+(e))$: if $ls(v_+(e)) = \mathbb{R}^d$, then it is the standard trivialization on $\mathbb{R}^d$; if $ls(v_+(e)) = \pi_i$ and $ls(> v_+(e)) = \emptyset$, then it is τ'_i ; if $ls(v_+(e)) \neq \mathbb{R}^d$ and $ls(> v_-(e)) = \{\pi_1, \pi_2\}$, then it is induced from the trivializations $\hat{\tau}'_i$ or τ'_i defined in Section ??.

Last, we glue this two pieces together to form $\mathbf{y}_{[e]}$. Notice that the result doesn’t depend on the choices of r, R , as long as they satisfy the required conditions: $R > R^{\min}(\mathbf{x}; v_+(e))$, $r < r^{\max}(\mathbf{x}; e)$, and $r/R = \epsilon$.

The above is the intuition of constructing $\mathbf{y}_{[e]}$. (Robin: end subsection here??)

For $T \in \mathcal{T}_A$ and any vertex $v \in T$, define

$$F(v) = \begin{cases} \mathbb{R}^d, & \text{if } ls(v) = \mathbb{R}^d \\ \text{the fiber of } ls(v) \text{ in which } \mathbf{x}_v \text{ lies,} & \text{otherwise.} \end{cases}$$

Definition 4.1 (Smoothing one node). Suppose $T \in \mathcal{T}(A)$ and $e \in E^c(T)$. Define

$$\psi_{T;e} : \text{dom}(T; e) \longrightarrow S_T \sqcup S_{T/e}$$

by first setting $\psi_{T;e}|_{S_T \times \{0\}}$ to be the projection $S_T \times \{0\} \rightarrow S_T$. For the remaining points in the domain, if $\mathbf{x} = (\mathbf{x}_v \in C_{Ch(v)}(ls(v)))_{v \in V(T)}$ and $\epsilon > 0$ is small enough so that $(\mathbf{x}, \epsilon) \in \text{dom}(T; e)$, then define

$$\psi_{T;e}(\mathbf{x}, \epsilon) = (y_v \in C_{Ch(v)}(ls(v)))_{v \in V(T/e)}$$

as follows. If v is not an endpoint of e , put $\mathbf{y}_{\text{contr}_{T;e}^V(v)} = \mathbf{x}_v$. It remains to define

$$\mathbf{y}_{[e]} \in C_{Ch([e])}(ls(v_-(e)) + ls(v_+(e)))$$

(Note that $Ch([e]) = Ch(v_+(e)) \sqcup Ch(v_-(e)) \setminus \{v_+(e)\}$.)

Throughout, we let $\star_e \in F(v_-(e))$ be the location of the point $v_+(e)$ in the configuration $\mathbf{x}_{v_-(e)}$ or, if $ls(v_-(e)) = \mathbb{R}^d$, the configuration $\mathbf{x}'_{v_-(e)}$. We define $\mathbf{y}_{[e]}$ via four cases:

- (a) If $ls(v_+(e)) = \mathbb{R}^d$ and $ls(v_-(e)) = \mathbb{R}^d$, then a representative of $\mathbf{y}_{[e]}$ is obtained from $\mathbf{x}'_{v_-(e)}$ by first removing $v_+(e) \in Ch(v_-(e))$ and then adjoining all points in $Ch(v_+(e))$ to $\mathbb{R}^d$, as the image of $\mathbf{x}'_{v_+(e)}$ under the map

$$\begin{aligned} (\mathbb{R}^d)^{Ch(v_+(e))} &\longrightarrow (\mathbb{R}^d)^{Ch(v_+(e))} \\ (w_v \in \mathbb{R}^d)_{v \in Ch(v_+(e))} &\longmapsto (\star_e + \epsilon \cdot w_v)_{v \in Ch(v_+(e))}. \end{aligned}$$

- (b) If $ls(v_+(e)) = \mathbb{R}^d$ and $ls(v_-(e)) \neq \mathbb{R}^d$, then $\mathbf{y}_{[e]}$ is obtained from $\mathbf{x}_{v_-(e)}$ by first removing $v_+(e) \in Ch(v_-(e))$ and then adjoining all points in $Ch(v_+(e))$ to E_e , as the image of $\mathbf{x}'_{v_+(e)}$ under the map

$$\begin{aligned} (\mathbb{R}^d)^{Ch(v_+(e))} &\longrightarrow (F(v_-(e)))^{Ch(v_+(e))} \\ (w_v \in \mathbb{R}^d)_{v \in Ch(v_+(e))} &\longmapsto (\exp_{\star_e}(\epsilon \cdot w_v))_{v \in Ch(v_+(e))}. \end{aligned}$$

- (c) If $ls(v_+(e)) \neq \mathbb{R}^d$ and $ls(v_-(e)) = \mathbb{R}^d$, then $\mathbf{y}_{[e]}$ is obtained from $\mathbf{x}_{v_+(e)}$ by adjoining all points in $Ch(v_-(e))$ except for $v_+(e)$, as the image of $\mathbf{x}'_{v_-(e)}$ under the map

$$\begin{aligned} (\mathbb{R}^d \setminus \mathbb{D}_{\star_e}^d(r^{\max}(x; e)))^{Ch(v_-(e))} &\longrightarrow (F(v_+(e)))^{Ch(v_-(e))}, \\ (w_v \in \mathbb{R}^d)_{v \in Ch(v_-(e))} &\longmapsto (\tau|_{F(v_+(e))})^{-1}(1/\epsilon \cdot (w_v - \star_e))_{v \in Ch(v_-(e))} \end{aligned}$$

where

$$\tau = \begin{cases} \tau'_i & \text{if } ls(v_+(e)) = \pi_i, i = 1 \text{ or } 2, \text{ and } ls(> v_+(e)) = \emptyset \\ \hat{\tau}'_i & \text{if } ls(v_+(e)) = \pi_i, i = 1 \text{ or } 2, \text{ and } ls(> v_+(e)) = \{\pi_{3-i}\} \\ \tau_{\bar{j}} & \text{if } ls(v_+(e)) = \pi_{\bar{j}}. \end{cases}$$

- (d) Suppose $ls(v_-(e)) = \pi_i$ and $ls(v_+(e)) = \pi_{3-i}$ for $i = 1$ or 2 .

- If $\star_e \in E_i(3)$, then $\mathbf{y}_{[e]}$, as a configuration of points in $\pi_{\bar{j}}$, lies on the fiber over $(\star_e, \pi_{3-i}(x_{v_+(e)}), \epsilon) \in E_i(3) \times B_{3-i} \times (0, \rho)$.
- If $\star_e \in E_i - \sigma_i(B_i) - E_i(2)$, write

$$\tau'_i(\star_e) = (\pi_i(x_{v_-(e)}), l, \theta) \in B_i \times \mathbb{R}^d \setminus \mathbb{D}^d(3),$$

where $(l \in [3, \infty), \theta \in S^{d-1})$ is the polar coordinate of the $\mathbb{R}^d \setminus \mathbb{D}^d(3)$ -factor. Then $\mathbf{y}_{[e]}$, as a configuration of points in $\pi_{\bar{j}}$, lies on the fiber over

$$\begin{aligned} ((-1)^{i-1}\theta, \pi_i(x_{v_-(e)}), \pi_{3-i}(x_{v_+(e)}), \gamma^{-1}(x, y)) &\in S^{d-1} \times B_i \times B_{3-i} \times (0, \rho) \times [-\frac{1}{2}, \frac{1}{2}] \\ &\approx S^{d-1} \times [-\frac{1}{2}, \frac{1}{2}] \times B_1 \times B_2 \times (0, \rho) \end{aligned}$$

where $x = 1/l, y = \epsilon$, and $\gamma : (0, \rho) \times [-1/2, 1/2] \rightarrow \mathbb{R}^2$ is the map in Section ??, defined right under the second picture in that section.

Recall in Section ?? we constructed $E_{\bar{j}}$ by doing surgery on E_1 and E_2 , so parts of $E_{\bar{j}}$ are naturally identified with parts of E_1 and E_2 . We define $\mathbf{y}_{[e]}$ as the union of all points in $\mathbf{x}_{v_-(e)}$ except $v_+(e)$ and all points $\mathbf{x}_{v_+(e)}$, which are viewed as points in $E_{\bar{j}}$ under this identification.

This completes the definition of $\psi_{T;e}$.

It is easy to be convinced that

$$\psi_{T;e}|_{\text{dom}(T;e)-S_T \times 0} : \text{dom}(T;e) - S_T \times 0 \rightarrow S_{T/e}$$

is smooth. (Maybe re-write the above in a better way to make this obvious.)

(Robin: Maybe make this a Proposition or Lemma.)

4.2 Smoothing multiple nodes (Robin's attempt to do almost all of them at once)

Given $T \in \mathcal{T}(A)$ with root r , write T as the following union of subtrees. First, if $ls(r) = \mathbb{R}^d$, let T_0 be the maximal subtree of T with root $r = r_0$ such that any vertex with space label other than $\mathbb{R}^d$ has no children in T_0 ; otherwise $T_0 = \emptyset$. Let T_1 be the maximal subtree whose root is the vertex $r_1 \neq r$ closest to r with $ls(r_1) \neq \mathbb{R}^d$ and such that any vertex with space label other than $\mathbb{R}^d$ has no children in T_1 . If there are two candidates for r_1 , we may choose r_1 so that $ls(r_1) = \pi_1$. It is possible that T_1 is a single vertex, and if $ls(r) = \pi_{\tilde{I}}$, then $T_1 = \emptyset$. Let T_2 be the maximal subtree whose root r_2 is the remaining vertex with $ls(r_2) \neq \mathbb{R}^d$; T_2 may be a single vertex. If some vertex is labeled by $\pi_{\tilde{I}}$, then T_2 is empty.

Now $T = T_0 \cup T_1 \cup T_2$, where each union is along a single vertex r_i such that $ls(r_i) \neq \mathbb{R}^d$.

Each T_i has an injection from a subset $A_i \subset A$ to its set of leaves, with $A = A_0 \sqcup A_1 \sqcup A_2$. Each T_i is not quite an A_i -tree because it is possible that only one of the space labels π_1 and π_2 appears in the tree. Nonetheless, we define S_{T_i} , with $i = 0, 1$, or 2 , in the same manner that we defined S_T for $T \in \mathcal{T}_A$.

For $i = 0, 1$, and 2 , we will define maps

$$S_{T_i} \times \prod_{e \in E^c(T_i)} [0, \epsilon_e) \rightarrow F(ls(r_i))^{A_i \sqcup R_i} \quad (1)$$

where $F(ls(r_i)) = \mathbb{R}^d$ if $ls(r_i) = \mathbb{R}^d$ and $F(ls(r_i))$ is the fiber of the bundle $ls(r_i)$ containing the configuration $\mathbf{x}_{r_i}$ otherwise, and where

$$R_i = \{r_j : j \in \{0, 1, 2\} \text{ and } r_j \text{ lies above } r_i \text{ but above no other } r_k\}.$$

A point in S_{T_i} is given by $(\mathbf{x}_v)_{v \in V(T_i)} \in \prod_{v \in V(T_i)} C_{Ch(v)}(ls(v))$. For $a \in A_i \sqcup R_i$, let $x_{v,a}$ be the point in the configuration $\mathbf{x}_v$ corresponding to the child of v that lies below the leaf a .

Writing $\star_{i,a} = x_{r_i,a}$ for $i = 0, 1$, or 2 , we define the map (1) by

$$\left((\mathbf{x}_v)_{v \in V(T_i)}, (t_e)_{e \in E^c(T_i)} \right) \mapsto \left(\exp_{\star_{i,a}} \left(\sum_{\substack{v < a \\ v \neq r_i}} \left(\prod_{\substack{e \in E^c(T_i) \\ v_+(e) < a}} t_e \right) x_{v,a} \right) \right)_{a \in A_i \sqcup R_i}.$$

Let T' be the tree obtained from T by contracting all the contractible edges in T_0 , T_1 , and T_2 . We have $0 \leq |E^c(T')| \leq 2$. If $|E^c(T')| = 0$, then we take ψ_T to be the map (1). If $|E^c(T')| > 0$, then each contractible edge in T' is directed either

- (i) from the root r with $ls(r) = \mathbb{R}^d$ to a vertex v with $ls(v) \neq \mathbb{R}^d$, or
- (ii) from a vertex with space label π_i to a vertex with space label π_{3-i} .

In cases (i) and (ii), we will apply the smoothing of a single node given in Definition 4.1, parts (c) and (d) respectively.

If $|E^c(T')| = 1$, this allows us to define ψ_T without ambiguity.

If $|E^c(T')| = 2$, we consider whether the order of contracting the two edges matters. In this case, either one or both edges is joined to the root. If only one is, the two smoothings affect disjoint disks in $ls(r)$; thus their composition can be expressed as a product of two maps and the order does not matter. If both contractible edges in T' are joined to the root, then the first contraction gives a smoothing as in Definition 4.1 (c) while the second gives a smoothing as in Definition 4.1 part (d). Regardless of the order of edge contractions, the leaves adjacent to the root in T' become (the same) configuration points in the equatorial region of E_I , so the order does not matter.

(Robin: Or maybe they're not exactly the same but related by a homotopy? We could also just fix one of the two orders, say by smoothing the edge to π_1 first.)

SKETCHY COMMENT: For the transition functions, we need only consider a map from a stratum to one of its substrata or vice-versa. Their smoothness should follow from the fact that the maps ψ_T are defined using exponential maps, trivializations at infinity, and the maps that glue the two bundles together, all of which are locally diffeomorphisms.

4.3 Smoothing out multiple nodes

(Robin: I left this version of smoothing multiple nodes that Xujia wrote, just in case.)

Given $T \in \mathcal{T}(A)$, and an (arbitrary) order on $E(T)$, we define $\psi_T : \text{dom}(T) \rightarrow \tilde{C}_A$ by smoothing out the nodes one by one. (From here to just above Lemma 4.2 can be skipped on first reading.)

(Robin: Doesn't Lemma 4.2 below mean that we don't need this itemized list?)

More precisely, given

$$((x_v)_{v \in V(T)}, (\epsilon_e)_{e \in E(T)}) \in \text{dom}(T) = S_T \times (\mathbb{R}^{\geq 0})^{E(T)},$$

$\psi_T((x_v), (\epsilon_e))$ is determined by the following inductive process: first, write

$$E'(T) = \{e \in E(T) \mid \epsilon_e \neq 0\} \subset E(T),$$

then:

- Start with $T_0 = T$, $E'(T)$, the induced order on $E'(T)$, and $X_0 = ((x_v)_{v \in T}, (\epsilon_e)_{e \in E'(T)})$.
- Given $T_i \in \mathcal{T}(A)$, a subset $E'(T_i) \subset E(T_i)$, an order on $E'(T_i)$, and

$$X_i = ((x_{i,v})_{v \in V(T_i)} \in S_{T_i}, (\epsilon_e \in \mathbb{R}^{>0})_{e \in E'(T_i)}),$$

we construct T_{i+1} , $E'(T_{i+1})$, an order on $E'(T_{i+1})$, and X_{i+1} :

- Let $e' \in E'(T_i)$ be the first element of $E'(T_i)$, then

$$T_{i+1} = T_i / e', \quad E'(T_{i+1}) = \text{contr}_{T_i; e'}^E(E'(T_i) \setminus \{e'\})$$

and the order on $E'(T_{i+1})$ is induced from the order on $E'(T_i)$ and $\text{contr}_{T_i; e'}^E$.

- For all $v \in V(T_i)$, $x_{i+1, \text{contr}_{T_i; e'}^V} = x_{i,v}$.
- $x_{i+1, [e']} = \psi_{T_i; e'}((x_{i,v})_{v \in V(T_i)}, \epsilon_{e'})$.
- For all $e \neq e' \in E'(T_i)$, if $\{v_-(e), v_+(e)\} \cap \{v_-(e'), v_+(e')\} = \emptyset$, then $\epsilon_{\text{contr}_{T_i; e'}^E(e)} = \epsilon_e$.
- It remains to determine $\epsilon_{\text{contr}_{T_i; e'}^E(e)}$ for other $e \neq e' \in E'(T_i)$:

- * If $ls(v_-(e')) = \mathbb{R}^d$, $ls(v_+(e')) \neq \mathbb{R}^d$, then, for $e \neq e' \in E(T_i)$:

$$\begin{cases} v_-(e) = v_+(e') \implies \epsilon_{\text{contr}_{T_i; e'}^E(e)} = \epsilon_e, \\ v_-(e) = v_-(e') \implies \epsilon_{\text{contr}_{T_i; e'}^E(e)} = \epsilon_e / \epsilon_{e'}, \\ v_+(e) = v_-(e') \implies \epsilon_{\text{contr}_{T_i; e'}^E(e)} = \epsilon_e \cdot \epsilon_{e'}. \end{cases}$$

- * If $ls(v_+(e')) = \mathbb{R}^d$, and either $ls(v_-(e')) \neq \mathbb{R}^d$ or

$$ls(v_-(e')) = \mathbb{R}^d, v_+(e) \text{ is not the out-most marked point in } x_{i, v_-(e)},$$

then, for $e \neq e' \in E(T_i)$:

$$\begin{cases} v_-(e) = v_+(e') \implies \epsilon_{\text{contr}_{T_i; e'}^E(e)} = \epsilon_e \cdot \epsilon_{e'}, \\ v_-(e) = v_-(e') \implies \epsilon_{\text{contr}_{T_i; e'}^E(e)} = \epsilon_e, \\ v_+(e) = v_-(e') \implies \epsilon_{\text{contr}_{T_i; e'}^E(e)} = \epsilon_e. \end{cases}$$

- * If $ls(v_-(e')) = ls(v_+(e')) = \mathbb{R}^d$ and $v_+(e)$ is the out-most marked point in $x_{i, v_-(e)}$, then... (not yet written; would be very annoying to write; you can probably see what I want to say; should find another way to write these, or change the second bullet in Definition 3.5)

- * If $ls(v_-(e')) = \pi_i$, $ls(v_+(e')) = \pi_{3-i}$: let $\star_e \in E_i$ be the position of the marked point $v_+(e')$ in the configuration $x_{i, v_-(e)}$.

- If $\star_e \in E_i(2)$, then, for all $e \neq e' \in E(T_i)$, $\epsilon_{\text{contr}_{T_i; e'}^E(e)} = \epsilon_e$.

- If $\star_e \in E_i - \sigma_i(E_i) - E_i(2)$: let $g_i : E_i - \sigma_i(B_i) \rightarrow \mathbb{R}^{\geq 1}$ be the function defined in Section ???. Then,

$$\begin{cases} v_-(e) = v_+(e') \implies \epsilon_{\text{contr}_{T_i; e'}^E(e)} = \epsilon_e, \\ v_-(e) = v_-(e') \implies \epsilon_{\text{contr}_{T_i; e'}^E(e)} = \epsilon_e \cdot g_i(\star_e) / 3, \\ v_+(e) = v_-(e') \implies \epsilon_{\text{contr}_{T_i; e'}^E(e)} = \epsilon_e \cdot 3 / g_i(\star_e). \end{cases}$$

- Define $\psi_T((x_v), (\epsilon_e)) = X_{|E'(T)|}$.

This completes the definition of ψ_T .

Lemma 4.2. ψ_T doesn't depend on the choice of order on $E(T)$.

Proof. (If I have not made a mistake (which I probably have...), this should be true and easy to verify. The definitions in Section 3.3 are designed for this lemma to hold.) $\square$

Corollary 4.3. For $T \neq T' \in \mathcal{T}(A)$, the transition map $\psi_{T'}^{-1} \circ \psi_T|_{\psi_T^{-1}(\text{image}(\psi_T) \cap \text{image}(\psi_{T'}))}$ is smooth. Therefore, $\tilde{C}_A$ is a well-defined smooth manifold with boundary and corners. For $T \in \mathcal{T}(A)$, $S_T \subset \tilde{C}_A$ is a codimension- $|E(T)|$ stratum.

If $A' \subset A$ is a finite subset, then we have an obviously-defined forgetful map $f_{A,A'} : \tilde{C}_A \rightarrow \tilde{C}_{A'}$.

Lemma 4.4. $f_{A,A'} : \tilde{C}_A \rightarrow \tilde{C}_{A'}$ is smooth. (proof not yet written)